

I. Single choice questions

1. The period of the autocorrelation sequence $R_{xx}[n]$ of the sequence $x[n] = (-1)^n$ is (**B**).
- A. 1 B. 2 C. 3 D. Aperiodic Signal
2. $X(e^{j\omega})$ is the DTFT of the real sequence $x[n]$. Define $Y(e^{j\omega}) = \text{Re}\{X(e^{j\omega})\}$, the IDTFT sequence $y[n]$ of $Y(e^{j\omega})$ is (**C**).
- A. $x[n] + x[-n]$ B. $x[n] - x[-n]$
C. $\frac{1}{2}(x[n] + x[-n])$ D. $\frac{1}{2}(x[n] - x[-n])$
3. The discrete-time system $y[n] = \sin(0.3\pi n) \sum_{m=0}^7 x[m-n]$ is (**A**).
- A. Linear time-varying B. Nonlinear time-varying
C. Linear time-invariant D. Nonlinear time-invariant
4. Among the following four sequences of length 9, the one whose 9-point DFT is a real sequence is (**D**), where $0 \leq n \leq 8$.
- A. $\{x_1[n]\} = \{4, 3, -5, 1, 2, -2, 1, -5, 3\}$ B. $\{x_2[n]\} = \{0, 5, 1, 4, -3, 3, -4, -1, -5\}$
C. $\{x_3[n]\} = \{0, -5, 2, 4, -3, 3, -4, -1, -5\}$ D. $\{x_4[n]\} = \{-5, 5, -2, 2, 4, 4, 2, -2, 5\}$
5. The unit impulse response of the system described by the difference equation $y[n] = \sum_{k=1}^{\infty} x(n-k)$ is (**A**).
- A. $u[n-1]$ B. $u[n]$ C. $u[n+1]$ D. non-existent
6. Regarding the Z-transform, which of the following statements is **correct**? (**BC**).
- A. The Z-transform of a two-sided sequence must exist.
B. If the ROC of a Z-transform includes the unit circle, the Fourier transform of

the corresponding time-domain sequence must converge.

C. If the Fourier transform of a time-domain sequence exists, its Z-transform's ROC must include the unit circle.

D. The ROC of a rational Z-transform may include poles.

7. If $H_1(z)$ and $H_2(z)$ are Type 2 and Type 4 linear-phase FIR filters respectively, then what type of linear-phase FIR filter is $H(z) = H_1(z)H_2(z)$? (**C**).

A. 1 B. 2 C. 3 D. 4

8. Any transfer function can be decomposed into the cascade of a minimum-phase transfer function and what type of transfer function? (**A**).

A. All-pass B. Maximum-phase C. Linear-phase D. Zero-phase

9. Given a filter with transfer function $G(z) = \frac{1}{2}(1 + z^{-5})$, which statement about this filter is incorrect? (**D**).

A. This is a Type 2 FIR linear-phase filter.

B. The filter has a notch at angular frequency π .

C. The filter has five zeros.

D. This is a fifth-order lowpass filter.

10. To design an IIR filter with a smooth passband magnitude response and an equiripple stopband magnitude response, one should employ (**D**).

A. the Elliptic filters

B. the Butterworth filters

C. the Type I Chebyshev filters

D. the Type II Chebyshev filters

II. Blank-fill questions

1. The discrete-time signal $x[n] = 5 \sin(0.4\pi n + \frac{\pi}{6})$ is obtained by sampling at a frequency of 2000 HZ. The frequency of its corresponding analog signal is_____.

Answer: 400Hz

2. Given the input-output relationship of a linear interpolator:

$y[n] = x[n] + \frac{2}{3}(x[n-1] + x[n+1]) + \frac{1}{3}(x[n-2] + x[n+2])$, the unit impulse response of this system is: _____

Answer: $h[n] = \{\frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3}\}, -2 \leq n \leq 2$

3. Given the discrete sequence $x[n] = \{ \underset{n=0}{\uparrow} 1, 5, 3, 8, 6 \}$, $x[< -n-3 >] =$ _____

Answer: $\{ \underset{\uparrow}{3}, 5, 1, 6, 8 \}$

4. Given the causal transfer function $H(z) = \frac{1}{1+0.6z^{-1}+0.05z^{-2}}$ and $G(z) = H(z/a)$, find the condition that a satisfies _____ for $G(z)$ to remain stable.

Answer: $|a| < 2$

5. Given a linear-phase FIR filter with phase function $\theta(\omega) = -2.5\omega + \pi/2$, this filter is a Type _____ linear-phase FIR filter.

Answer: IV (or 4)

6. Determine the ~~4~~-point circular convolution $y_c[n]$ of the sequences $g[n] = \{2, 0, 2, 5\}, 0 \leq n \leq 3$ and $h[n] = \{6, 2, 3\}, 0 \leq n \leq 2$, so $y_c[n]$ is _____

Answer: $\{27, 4, 18, 34\}, 0 \leq n \leq 3$

7. The computational complexity of the complex multiplication operations in an N-point sequence FFT is _____

Answer: $(N \log_2 N)/2$

8. The causal stable system that has the same magnitude response as the inverse system of the transfer function $G(z) = \frac{(z-4)(z+5)}{(z+0.5)(z-0.3)}$ is _____

Answer: $G_{\min}^{-1}(z) = \frac{(z+0.5)(z-0.3)}{(4z-1)(5z+1)}$

9. The causal IIR digital transfer function $G(z) = \frac{1+z^{-1}}{2-z^{-1}}$ was designed using the bilinear transformation method with $T=1$. Find its causal analog transfer function $H_a(s) = \underline{\hspace{2cm}}$, and determine that this digital filter has a
 $\underline{\hspace{2cm}}$ (low-pass, high-pass, band-pass, band-stop) magnitude frequency response characteristic.

Answer: $H_a(s) = G(z) \Big|_{z=\frac{2+s}{2-s}} = \frac{z+1}{2z-1} \Big|_{z=\frac{2+s}{2-s}} = \frac{4}{3s+2}$, low-pass

III. Calculations

1. Given the sequence $x(n) = j \cos(2\pi n / 7) - \sin(2\pi n / 3)$,

whose DTFT is $X(e^{j\omega})$:

- 1) Is this sequence periodic? If yes, determine its period.
- 2) Write the conjugate-symmetric part $x_{cs}(n)$ and conjugate-antisymmetric part $x_{ca}(n)$ of the sequence. Express their DTFTs in terms of $X(e^{j\omega})$.

Answer: 1) $N_1 = 2\pi / (2\pi / 7) = 7$; $N_2 = 2\pi / (2\pi / 3) = 3$

It's periodic, and $N = 21$

2) $x^*(n) = -j \cos(2\pi n / 7) - \sin(2\pi n / 3)$

$$x^*(-n) = -j \cos(2\pi n / 7) + \sin(2\pi n / 3)$$

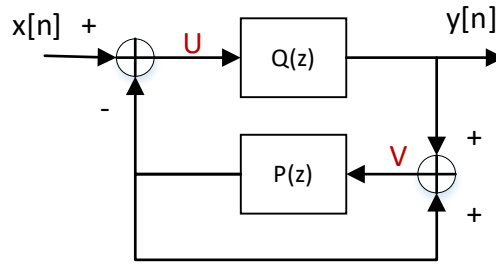
$$x_{cs}(n) = \frac{x(n) + x^*(-n)}{2} = 0$$

$$x_{ca}(n) = \frac{x(n) - x^*(-n)}{2} = j \cos(2\pi n / 7) - \sin(2\pi n / 3)$$

$$x_{cs}(n) \leftrightarrow X_{re}(e^{j\omega}) = 0$$

$$x_{ca}(n) \leftrightarrow jX_{im}[k] = X(e^{j\omega})$$

2. For a LTI discrete-time system whose structure is shown in the figure, determine its inverse system.



Answer: Set $U(z)$ and $V(z)$, then

$$U(z) = X(z) - P(z)V(z)$$

$$V(z) = Y(z) + P(z)V(z)$$

$$Y(z) = Q(z)U(z)$$

$$H(z) = Y(z)/X(z) = \frac{Q(z)(1 - P(z))}{1 - P(z) + P(z)Q(z)}$$

The inverse system is

$$G(z) = \frac{1}{H(z)} = \frac{1 - P(z) + P(z)Q(z)}{Q(z)(1 - P(z))}$$

3. Let $x_1[n]$ and $x_2[n]$ be two real-valued sequences of length N . A complex-valued sequence is formed as $x[n] = x_1[n] + jx_2[n]$, whose N -point DFT is denoted by $X[k]$. Based on $x_1[n]$ and $x_2[n]$, construct a new real-valued sequence $y[n]$ of length $2N$ such that:

$$\begin{aligned} y[2n] &= x_1[n] \\ y[2n+1] &= x_2[n], \quad 0 \leq n \leq N-1 \end{aligned}$$

Express the $2N$ -point DFT of $y[n]$ in terms of $X[k]$.

Answer:

$$X_1[k] = \frac{1}{2} \{X[k] + X^*[\langle -k \rangle_N]\},$$

$$X_2[k] = \frac{1}{2j} \{X[k] - X^*[\langle -k \rangle_N]\}$$

$$\begin{aligned}
Y[k] &= X_1[\langle k \rangle_N] + W_{2N}^k X_2[\langle k \rangle_N] \\
&= \frac{1}{2} \{X[\langle k \rangle_N] + X^*[\langle -k \rangle_N]\} + \frac{1}{2j} W_{2N}^k \{X[\langle k \rangle_N] - X^*[\langle -k \rangle_N]\} \\
&= \frac{1}{2} (1 - jW_{2N}^k) X[\langle k \rangle_N] + \frac{1}{2} (1 + jW_{2N}^k) X^*[\langle -k \rangle_N]
\end{aligned}$$

4. Determine the lowest-order causal real-coefficient allpass transfer function with two poles at $p_1 = 0.4$ and $p_2 = 0.3 + j0.4$.

Answer: According the real-coefficient, the third pole is

$$p_3 = p_2^* = 0.3 - j0.4$$

$$\begin{aligned}
\text{then:} \quad D(z) &= (1 - p_1 z^{-1})(1 - p_2 z^{-1})(1 - p_3 z^{-1}) \\
&= (1 - 0.4z^{-1})(1 - 0.6z^{-1} + 0.25z^{-2}) \\
&= 1 - z^{-1} + 0.49z^{-2} - 0.1z^{-3}
\end{aligned}$$

$$\text{Thus} \quad A_3(z) = \frac{-0.1 + 0.49z^{-1} - z^{-2} + z^{-3}}{1 - z^{-1} + 0.49z^{-2} - 0.1z^{-3}}$$

5. A FIR filter of length 3 is defined with an impulse response given by $h[0] = h[2]$.

When the sum of two cosine sequences $\cos(0.25\pi n)$ and $\cos(0.75\pi n)$ is input to this filter, the output contains only the high-frequency complement.

- 1) Determine the filter's impulse response $h[n]$.
- 2) Determine whether the system is linear phase.

Answer:

$$1) \quad h[n] = 1 + az^{-1} + z^{-2}$$

$$\text{There is a zero at } z = e^{j0.25\pi}$$

$$1 + ae^{-j0.25\pi} + e^{-2j0.25\pi} = 0$$

$$\text{So } a = -\sqrt{2}$$

$$h[n] = \{1, -\sqrt{2}, 1\}$$

- 2) It is linear phase system

6. The specifications for a linear-phase FIR bandstop filter are given as:

Sampling frequency: $F_T=40\text{kHz}$

Passband cutoff frequencies: $F_{p1}=9\text{kHz}$, $F_{p2}=15\text{kHz}$

Stopband cutoff frequencies: $F_{s1}=10\text{kHz}$, $F_{s2}=13\text{kHz}$

All passband and stopband ripples are constrained to be less than 0.001.

Using the window functions and their characteristics provided in the Appendix, design a linear-phase FIR filter with minimum length that satisfies these specifications.

Answer: 1)

$$\alpha_s = -20 \log_{10}(0.001) = 60 \text{ dB}$$

$$\omega_{s1} = 2\pi \frac{F_{s1}}{F_T} = 0.5\pi, \quad \omega_{s2} = 2\pi \frac{F_{s2}}{F_T} = 0.65\pi,$$

$$\omega_{p1} = 2\pi \frac{F_{p1}}{F_T} = 0.45\pi, \quad \omega_{p2} = 2\pi \frac{F_{p2}}{F_T} = 0.75\pi$$

2) Choose **Blackman**

$$w[n] = 0.42 + 0.5 \cos\left(\frac{2\pi n}{2M+1}\right) + 0.08 \cos\left(\frac{4\pi n}{2M+1}\right), -M \leq n \leq M$$

$$(3) \quad \Delta\omega_1 = 0.05\pi, \quad \Delta\omega_2 = 0.1\pi$$

$$5.56\pi/M \leq 0.05\pi$$

$$M \geq 111.2$$

$$M=112$$

(4)

$$\omega_c = (\omega_s + \omega_p) / 2$$

$$\omega_{c1} = 0.475\pi, \quad \omega_{c2} = 0.7\pi$$

$$h_{BS}[n] = \begin{cases} 1 - \frac{\omega_{c2} - \omega_{c1}}{\pi} = 0.775 & n = 0 \\ \frac{\sin(0.475\pi n)}{\pi n} - \frac{\sin(0.7\pi n)}{\pi n} & |n| > 0 \end{cases}$$

(5)

$$h_d[n] = h_{BS}[n]w[n]$$

(6)

$$h[n] = h_d[n-M]w[n-M], \quad 0 \leq n \leq 2M$$

Appendix:**Table1 Some fixed window functions**

Rectangular:	$w[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$
hann:	$w[n] = \frac{1}{2} [1 + \cos(\frac{2\pi n}{2M+1})], -M \leq n \leq M$
hamming:	$w[n] = 0.54 + 0.46 \cos(\frac{2\pi n}{2M+1}), -M \leq n \leq M$
blackman:	$w[n] = 0.42 + 0.5 \cos(\frac{2\pi n}{2M+1}) + 0.08 \cos(\frac{4\pi n}{2M+1}), -M \leq n \leq M$

Table 2 Properties of some fixed window functions

Type of window	Main lobe width Δ_{ML}	Relative Sidelobe Level A_{sl}	Minimum Stopband Attenuation	Transition Bandwidth $\Delta\omega$
Rectangular	$4\pi/(2M+1)$	13.3dB	20.9dB	$0.92\pi/M$
Hann	$8\pi/(2M+1)$	31.5dB	43.9dB	$3.11\pi/M$
Hamming	$8\pi/(2M+1)$	42.7dB	54.5dB	$3.32\pi/M$
Blackman	$12\pi/(2M+1)$	58.1dB	75.3dB	$5.56\pi/M$